



Sir William Tritton beside his bridge-carrying tractor

Great designers Tritton and Wilson

After years of stalemate on the Western Front, the Allies finally pierced the German lines in 1917, using an invention that dismayed the enemy—the tank. Designed and constructed in secret, it was the brainchild of two British engineers, William Tritton and Walter Wilson.

WILLIAM TRITTON JOINED the agricultural machinery manufacturers Fosters of Lincoln in 1905 as general manager. He had a background in pump manufacturing and general engineering, and helped Fosters promote its new general purpose agricultural tractor in South America. While at Fosters, he worked with David Roberts in creating a tracked engine to be exported to the Yukon in Canada. Roberts later sold his track patents to the Holt company in the US.

Tritton also promoted the gasoline engine tractor, and just before World War I, both a 40bhp and a 105 bhp Foster Daimler tractor were marketed. The outbreak of war brought an order for 97 of the huge new tractors to pull naval siege guns. One of the tractors, OHMS No. 44, was adapted to



Sir William Tritton
(1875-1946)



Walter Wilson
(1874-1957)

carry a 15 ft (4.5 m) bridge, which was slung under its main framework and pushed forward to cross a trench. The experiment was abandoned, but Fosters became known to the authorities for their innovation and speedy vehicles.

In February 1915, Winston Churchill established the Landships Committee to promote mechanical answers to the stalemate on the Western Front. Fosters was approached to start work on one of the early ideas—a big wheel to roll across the fields of barbed wire that protected the German trenches. This project was soon canceled, but in July 1915, following demonstration of a Holt tractor, the Ministry of Munitions placed an order with Fosters for an experimental tracked armored machine. Design work began on August 2, building began on August 11, and the vehicle was first driven on September 8—an extraordinary speed of manufacture by anyone's standards. Only in late August did Tritton hear from the War Office that the machine should be capable of crossing a 5 ft (1.5 m) wide trench and mounting a 4½ ft (1.4 m) parapet, which was beyond its capabilities. As work continued on the No. 1 Lincoln machine (or "Little Willie" as she was later called), Lieutenant Walter Wilson began building a new vehicle with Tritton's assistance. Walter Wilson, a Royal Naval Volunteer Reserve officer, had designed a



A top-secret design

Tritton's prototype tank, "Little Willie," was kept under wraps during trials to maintain secrecy. It was the first ever completed tank prototype.

car and a truck before the war. On joining the Royal Naval Air Service team that was working on solutions to trench warfare, he realized that there was a problem with Little Willie's shape. His solution was the new rhomboidal design that is now so familiar—that of the classic World War I tank, complete with tracks looped around its entire body. He also designed the sponsons that housed the tank's guns. On September 26, a wooden model of the tank was approved, and the new prototype, called "Mother," was built in just 99 days.

Wilson was sent to the Metropolitan Carriage and Wagon Company, near Birmingham, to supervise the manufacture of Mark I tanks—125 of which were ordered from Metropolitan and 25 from Fosters, which had a much smaller manufacturing capacity. At Birmingham, Wilson continued designing and was influential in having the Ricardo engine approved for the Mark V tank. This had Wilson's own new gearbox, which enabled the tank to be driven by one man.

Tritton, meanwhile, had started design on a new, faster tank called the Tritton Chaser, which was accepted into service as the Medium Mark A tank, or Whippet. The Chaser had two Tylor engines, one to power each track, and was intended as a



Mark IV tank at the front
Canadian troops pose atop a Mark IV tank in 1918. The wooden beam was placed under the vehicle's tracks when it was stuck in mud.